

Market Concentration Case Studies

Browser Industry and Credit-Card Issuer Analysis Using HHI and CR4

Executive Summary

This portfolio artifact applies Industrial Organization concepts to two market-structure case studies: web browsers and credit-card issuers. The analysis uses concentration metrics such as the four-firm concentration ratio (CR4) and the Herfindahl-Hirschman Index (HHI) to compare how different measures can tell different stories about competition and firm dominance.

The browser case shows why HHI can be more informative than a simple top-four concentration ratio when one firm becomes increasingly dominant inside an already concentrated market. The credit-card issuer case shows how a merger-style scenario can be evaluated by combining market shares and recomputing HHI.

Method

The Herfindahl-Hirschman Index is calculated by summing the squares of firm market shares. When shares are expressed as percentages, HHI is typically presented on a 0–10,000 scale. CR4 is the combined market share of the four largest firms. HHI is more sensitive to dominance by the largest firm because squaring gives greater weight to larger shares.

Metric	Formula / logic	What it helps show
CR4	Sum of the four largest firm shares	How much of the market is held by leading firms.
HHI	Sum of squared firm shares	Both overall concentration and dominance inside the market.
Delta HHI	Post-scenario HHI minus base HHI	How concentration changes after a merger or consolidation scenario.

These case studies are intended as educational portfolio demonstrations of transparent economic reasoning, spreadsheet-based concentration analysis, and careful interpretation of what common concentration metrics can and cannot show.

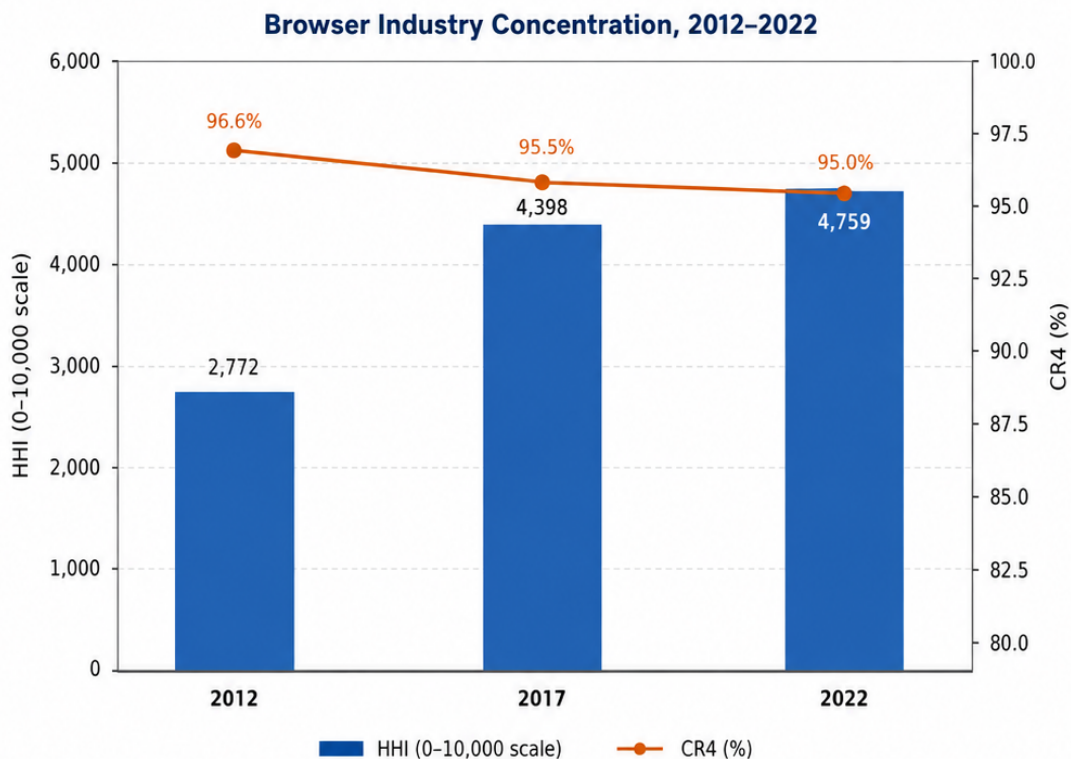
Case 1: Browser Industry

The browser case compares market-share snapshots from 2012, 2017, and 2022. Browser shares are aggregated to the firm level before calculating HHI; for example, Microsoft browser shares are treated as Microsoft share rather than separate firms.

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Metric	2012	2017	2022
HHI	2,772	4,398	4,759
CR4	96.6%	95.5%	95.0%
Largest firm share	32.8%	63.2%	66.9%



HHI values are on a 0–10,000 scale (higher values indicate greater concentration).

Figure: Browser industry concentration summary using HHI and CR4 across selected years.

Browser Interpretation

The top-four share remains very high across the period, but the HHI rises sharply. That difference matters: CR4 shows that a small number of firms remain dominant, while HHI shows that concentration within that leading group increases as Google Chrome's share expands.

Case 2: Credit-Card Issuer Concentration

The credit-card issuer case uses issuer market shares to calculate baseline concentration and then models a scenario in which Capital One and Discover are combined. This demonstrates how HHI changes when two firm shares are consolidated into a single entity.

Metric	Base	Hypothetical scenario	Change
HHI - named issuers	886	1,053	+168
CR4 - base	51.0%	-	-
CR5 - base	60.9%	-	-

Interpretation

The issuer case is less concentrated than the browser case in HHI terms, but the scenario still illustrates a meaningful increase in concentration. The exercise is useful because it shows how market-share arithmetic translates into a structured concentration analysis.

A concentration metric is a starting point, not a legal conclusion. In practice, any merger or competition assessment would also require broader evidence about market definition, entry barriers, consumer switching costs, pricing behavior, innovation, and firm conduct.

Portfolio Relevance

This project demonstrates applied microeconomic reasoning, spreadsheet-based data analysis, and policy-relevant interpretation. It shows how an economic measure can be calculated transparently, checked against alternative metrics, and translated into a careful narrative about market structure.

- Industrial Organization and antitrust-style economic reasoning
- Market-share data organization and firm-level aggregation
- HHI and CR4 calculation
- Scenario analysis using recomputed firm shares
- Interpretation of what concentration metrics can and cannot show
- Professional communication of quantitative findings

Caution

These case studies are portfolio demonstrations of economic analysis. They should not be read as legal conclusions about market power, consumer harm, or merger enforcement. A full competition analysis would require broader evidence about market definition, entry barriers, pricing behavior, innovation, switching costs, and firm conduct.